

1. Fill in the blanks with appropriate answers.

	Speed in kmph	Speed in m/s
(a)	18	?
(b)	36	?
(c)	?	15
(d)	63	?
(e)	?	20
(f)	?	25
(g)	108	?

2. A person travelled from city A to city B at a speed of 15 m/sec in four hours. What is the distance between the two cities?

- (1) 176 km
(2) 196 km
(3) 216 km
(4) 236 km
(5) None of these

3. A car travels 600 km in 15 hours. If the speed is increased by 10 km/hr, then how much time does it take to cover the same distance?

- (1) 13 hours
(2) $12\frac{1}{2}$ hours
(3) 12 hours
(4) 10 hours
(5) None of these

4. A and B start together from city P to city Q with respective speeds in the ratio 3 : 2. If B takes $2\frac{1}{2}$ hours more than A to reach Q, then how much time does A take to reach Q?

- (1) 5 hours
(2) 4 hours
(3) $7\frac{1}{2}$ hours
(4) 10 hours
(5) None of these

5. Walking at $\frac{3}{4}$ th of his usual speed, a man reaches his office 20 minutes late. Find the usual time he takes to reach the office.

- (1) 40 min
(2) 45 min
(3) 50 min
(4) 60 min
(5) None of these

6. A worker reaches his office 10 minutes late, walking at 6 kmph. He reaches his office 5 minutes early, walking at 8 kmph. Find the distance between his house and office.

- (1) 4 km
(2) 4.8 km
(3) 6 km
(4) 7.2 km
(5) None of these

7. Sunny travelled a distance of 500 km for 9 hours. He covered a certain part at a speed of 60 kmph and the remaining part of the distance at a speed of 50 kmph. What is the time taken by him for the distance covered at a speed of 50 kmph?

- (1) 5 hours
(2) 4 hours
(3) 6 hours
(4) 7 hours
(5) 3 hours

8. A man covers a certain distance travelling at a speed of 40 kmph and returns to the starting point at a speed of 60 kmph. Find the average speed for the entire journey.

- (1) 50 kmph
(2) 48 kmph
(3) 54 kmph
(4) 52 kmph
(5) None of these

9. Charan travels from city X to city Y, a distance of 280 km, at a speed of 56 kmph. He travels from city Y to city Z, a distance of 320 km, at a speed of 40 kmph and from city Z to city X, a distance of 150 km, at 75 kmph. What is the average speed of Charan for the whole journey?

- (1) 45 kmph
(2) 50 kmph
(3) 60 kmph
(4) 75 kmph
(5) None of these

10. Bunny travelled from city A to city B at a speed of 45 kmph and from city B to city C at a speed of 54 kmph. If the distance from city A to city B and that from city B to city C are in the ratio 5 : 4 respectively, then what is the average speed of Bunny over the entire journey?

- (1) 46 kmph
(2) 48 kmph
(3) 49.5 kmph
(4) 50 kmph
(5) None of these

11. Satish and Sudhir started running from a point in opposite directions at speeds of 8 kmph and 10 kmph respectively. What would be the distance between them after four hours and 30 minutes?

- (1) 54 km
(2) 63 km
(3) 72 km
(4) 81 km
(5) None of these

12. A thief who is at a distance of 500 m from a policeman started moving away on his bike at a speed of 54 kmph. Observing this, the policeman too started at the same time at 72 kmph to chase the thief. How much time will the police man take to catch the thief?

- (1) 100 sec
(2) 150 sec
(3) 200 sec
(4) 250 sec
(5) None of these

13. Saha, travelling at a speed of 84 kmph, is six km behind Nihal. After 30 minutes, Saha will be six km ahead of Nihal. Find the speed of Nihal.

- (1) 72 kmph
- (2) 60 kmph
- (3) 90 kmph
- (4) 66 kmph
- (5) None of these

14. At 10:00 a.m. Swetha started from town M and travelled towards N at 40 kmph and at 11:15 a.m. Swathi started from town M and travelled to town N at a speed of 50 kmph. At what time would Swathi overtake Swetha?

- (1) 4:15 p.m.
- (2) 5:30 p.m.
- (3) 3:00 p.m.
- (4) 3:45 p.m.
- (5) None of these

15. Two cities P and Q are separated by a distance of 450 km. Train A starts from city P to reach city Q five hours after train B starts from city P. Train A runs 15 kmph faster than train B. Both the trains reach city Q at the same time. What is the speed of train A?

- (1) 45 kmph
- (2) 50 kmph
- (3) 60 kmph
- (4) 75 kmph
- (5) None of these

16. A train of length 500 m crosses an electric pole in 25 seconds. The train crosses a platform traveling at the same speed in 49 seconds. What is the length of the platform?

- (1) 450 m
- (2) 480 m
- (3) 500 m
- (4) 540 m
- (5) None of these

17. Find the time taken by a 150 m long train running at 36 km/hr to cross another train of length 200 m and running at 54 km/hr in the

(i) same direction.

- (1) 50 seconds
- (2) 60 seconds
- (3) 70 seconds
- (4) 80 seconds
- (5) None of these

(ii) opposite direction.

- (1) 10 seconds
- (2) 16 seconds
- (3) 8 seconds
- (4) 14 seconds
- (5) None of these

18. A person riding on his bike at a speed of 72 kmph crossed a train moving in the same direction at a speed of 60 kmph in two minutes. What is the length of the train?

- (1) 500 m
- (2) 450 m
- (3) 400 m
- (4) 350 m
- (5) None of these

19. Two trains of lengths 200 m and 250 m run on parallel tracks. When they run in the same direction, it will take 25 seconds to cross each other and when they run in opposite direction it will take 15 seconds to cross each other. Find the speed of the faster train.

- (1) 6 m/sec
- (2) 12 m/sec
- (3) 18 m/sec
- (4) 24 m/sec
- (5) None of these

20. Two trains of equal lengths are running at uniform speeds of 63 kmph and 45 kmph respectively. The two trains crossed each other in 45 seconds when travelling in opposite directions. In how much time would the faster train overtake the slower train?

- (1) 5 minutes
- (2) $4\frac{1}{2}$ minutes
- (3) $5\frac{2}{3}$ minutes
- (4) $6\frac{1}{2}$ minutes
- (5) None of these

Directions for questions 1 to 20: Select the correct alternative from the given choices.

1. A train can cross a 450 m long bridge in one minute and 20 seconds, whereas it can cross a 750 m long bridge in one minute 40 seconds. What is the length of the train?
(1) 400 m (2) 450 m
(3) 500 m (4) 750 m
(5) None of these
2. A bus was running at a speed of 45 kmph and the speed of the bus increased by 3 kmph at the end of every hour. What is the total distance covered by the bus in the first 12 hours of the journey?
(1) 750 km (2) 762 km
(3) 735 km (4) 756 km
(5) None of these
3. M and N are two stations which are 918 km apart. A train starts from station M at 8:45 a.m. and travels at a speed of 72 kmph towards station N. Another train starts from station N at 11:00 a.m. and travels at a speed of 54 kmph towards station M. At what time would the two trains cross each other?
(1) 7:00 p.m. (2) 7:15 p.m.
(3) 2:45 p.m. (4) 5:00 p.m.
(5) None of these
4. Train P crosses a stationary train Q in 54 seconds and a bridge of length 1100 m in 72 seconds travelling at the same speed. The length of train P is 900 m. What is the length of train Q?
(1) 400 m (2) 500 m
(3) 600 m (4) 700 m
(5) None of these
5. Excluding the stoppages, the average speed of a train is 120 kmph and including the stoppages, the average speed of the train is 90 kmph. For how many minutes does the train stop per hour?
(1) 15 minutes (2) 12 minutes
(3) 18 minutes (4) 13 minutes
(5) None of these
6. A man swims 20 km in 2 hours along the stream and 12 km in 2 hours against the stream. Find the speed of the stream.
(1) 1.5 kmph (2) 4 kmph
(3) 2 kmph (4) 3 kmph
(5) None of these
7. A person can row a boat at 12 kmph in still water. He takes seven hours and 28 minutes to row from X to Y and return to X. If the speed of the stream is 3 kmph, what is the distance between points X and Y?
(1) 42 km (2) 45 km
(3) 48 km (4) 39 km
(5) None of these
8. A man rowing a boat takes two hours 24 minutes more to travel 54 km upstream than to travel the same distance downstream. If the speed of the stream is 3 kmph, what is the speed of the boat in still water?
(1) 13.5 kmph (2) 12 kmph
(3) 15 kmph (4) 10.5 kmph
(5) None of these
9. A boat can travel 6.4 km upstream in 48 minutes. If the ratio of the speed of the stream to the speed of the boat in still water is 1 : 5, how much time will the boat take to cover 7.2 km down stream?
(1) 24 minutes (2) 48 minutes
(3) 54 minutes (4) 36 minutes
(5) None of these
10. The speed of a boat in still water is five times the speed of the stream. It took four hours 30 minutes to cover a distance of 54 km of downstream journey long. How much time would it take to cover a 30 km upstream journey?
(1) 4 hr 30 minutes (2) 3 hr 12 minutes
(3) 3 hr 45 minutes (4) 4 hr 48 minutes
(5) None of these
11. A man can row at 10 kmph in still water. If the speed of the water current is 2 km/hr and if it takes 5 hours for him to go to a new place and return from there, then find the distance between his starting point and the new place.
(1) 24 km (2) 48 km
(3) 16 km (4) 20 km
(5) None of these
12. In a 100 m race, A beats B by 10 m and B beats C by 10 m. By how many meters will A beat C?
(1) 21 m (2) 19 m
(3) 16 m (4) 17 m
(5) None of these
13. In a 100 m race, A beats B by 10 m or by 2 seconds. Find the speed of A.
(1) 18 kmph (2) 15 kmph
(3) 20 kmph (4) 16 kmph
(5) None of these
14. In a kilometer race, A beats B by 25 seconds and B beats C by 15 seconds while A beats C by 200 m. What is the time taken by B to complete the race?
(1) 3 min 5 sec (2) 2 min 45 sec
(3) 3 min 25 sec (4) 4 min 10 sec
(5) None of these

15. In a kilometer race, P beats Q by 200 m and beats R by 424 m. Find the distance by which Q beats R in a 1500 m race (in m).

(1) 240 (2) 480
(3) 210 (4) 420
(5) None of these

16. P, Q, R and S ran a 900 m race. P beats Q by 180 m, Q beats R by 112.5 m and S beats R by 200 m. Which of the following can be concluded?

(1) P beats S by 100 m (2) P beats S by 90 m
(3) S beats P by 100 m (4) S beats P by 90 m
(5) None of these

17. In a 400 m circular track, A and B start from the same point and at the same time with the speeds of 12 km/hr and 18 km/hr respectively. When will they meet again for the first time on the track, if they are

(i) running in the same direction.
(ii) running in the opposite direction.

(1) 160. sec, 24 sec
(2) 200 sec, 48 sec
(3) 240 sec, 24 sec
(4) 240 sec, 48 sec
(5) None of these

18. Three persons A, B and C run around a circular track of length 600 m with the speeds of 10 km/hr, 12 km/hr and 18 km/hr respectively. If they start at

the same point and at the same time and move in the same direction, then find

(i) when will they meet for the first time at the starting point.
(ii) the time taken to meet for the first time anywhere on the track.

(1) 1080 sec, 1080 sec
(2) 320 sec, 160 sec
(3) 960 sec, 1280 sec
(4) 320 sec, 150 sec
(5) None of these

19. Two persons X and Y started running simultaneously from a point around a 900 m long circular track with speeds of 12 m/sec and 18 m/sec respectively in opposite directions. Find the time taken by them to meet for the first time (in seconds).

(1) 20 (2) 30 (3) 24
(4) 18 (5) None of these

20. A, B and C started running simultaneously from a point on a circular track. The speed of C was half of that of B and thrice that of A. A took 12 minutes to complete a round. Find the time taken by all the three to meet at the starting point for the first time (in minutes).

(1) 6 (2) 18 (3) 24
(4) 12 (5) None of these